

PREDICTING FINANCIAL DISTRESS IN US RETAILERS FROM SEC XBRL DISCLOSURES

TEAM 18

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INPUT & OUTPUT SPECIFICATIONS

INPUT SPECIFICATION

- **16 Raw SEC XBRL Financial Items:** Quarterly 10-K & 10-Q balance sheets and income statements across 8 retail chains, enforcing strict point-in-time timestamping (available_on $\leq$ as_of).
- **47 Multi-Window Engineered Ratios:** Derived across rolling windows (≤ 8 quarters): solvency, liquidity, profit margins, capital leverage, YoY changes, and historical min/max values.
- **Strict Out-of-Scope Boundaries:** Excludes contemporaneous market stock prices, macroeconomic indices, and credit agency ratings; strictly prohibits one-hot ticker encoding shortcuts.

OUTPUT SPECIFICATION

- **Binary Risk Classification:** Supervised label $Y \in \{0, 1\}$: 0 = Sound / Healthy; 1 = Impairment / Financial Distress in the subsequent forward quarter.
- **Calibrated Probability & Audit Priority:** Continuous posterior $P(\text{distress}) \in [0, 1]$ mapped to operational threshold $t = 0.7879$ ($C(\text{FN}) = 5 \cdot C(\text{FP})$) for audit escalation.
- **Local Profile Explainability (SHAP):** Instance-level KernelSHAP feature attributions quantifying exact marginal contributions of liquidity, leverage, and cash burn toward each risk alert.



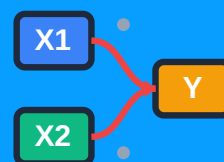
Corpus Architecture & Purged Time-Split



SEC Filings Universe

332 Quarters

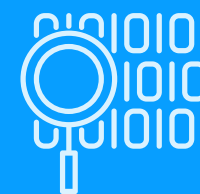
DG, DKS, FIVE, HD, LOW,
ORLY, ROST, WMT



Observation Matrix

324 Samples

8-quarter temporal
sliding windows



Audited Cell Matrix

4,609 Data Cells

0 SHA-256
discrepancies



TRAINING SET

212

- 62.3% Distress Positive
- In-sample parameter tuning



PURGED BUFFER

16

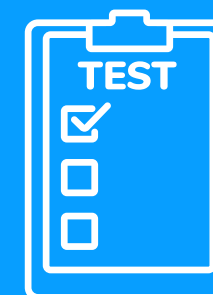
- 90-Day Quarantine
- IFiling latency buffer



VALIDATION

32

- 65.6% Distress Positive
- Threshold selection



LOCKED TEST

64

- 59.38% Distress Positive
- Strict out-of-time test

DUAL-LEVEL IMBALANCE PARADOX

LEVEL 1 • SAMPLE OBSERVATIONS

MODERATE SAMPLE IMBALANCE

Aggregate observation splits appear benign across standard class balance diagnostics.

- Imbalance Ratio: 1.66
- Distress Ratio: 62.35%
- Shannon Entropy: 0.9556 bit

High Shannon entropy falsely suggests a balanced learning problem to naive estimators.

LEVEL 2 • ENTITY CONCENTRATION

EXTREME ENTITY DISTORTION

Individual retail chains exhibit deterministic, single-class regime persistence.

HD 100% Distressed	LOW 100% Distressed	WMT 100% Distressed	ROST 4.7% Distressed
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41.0% of total corpus originates from purely single-class firms, creating an entity memorization vulnerability.

Feature Signals & Multicollinearity

TOP PREDICTIVE FEATURE

current_ratio_min_window

|2·AUC – 1|

0.904

Normalized Gini

Raw AUC

0.048

Inverse rank signal

Mutual Info

0.432

MI metric

Inverse Ranking: Raw AUC = 0.048 indicates strong separation where lower liquidity scores monotonically predict imminent distress events.

MULTICOLLINEARITY AUDIT

Extreme Feature Redundancy

Severe Collinearity

33 / 47

Features VIF > 10

Peak Inflation

798.10

Maximum VIF observed

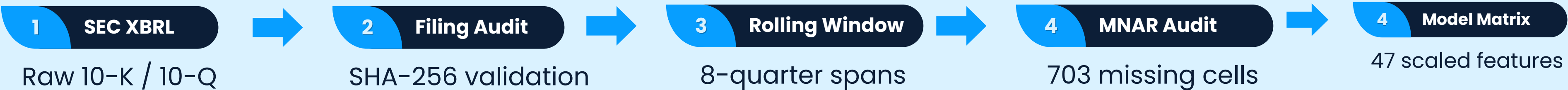
Effective Dim

12.89

Participation ratio

Dimensionality Collapse: 47 nominal accounting metrics span less than 13 independent linear coordinates.

Data Preprocessing & MNAR Mechanism



MNAR DISCLOSURE ARTIFACT

703 Raw Missing Values

Missing Not At Random mechanism drives reporting omissions across SEC filings.

Net Margin Disclosure Omission

Healthy Firm Correlation

Distress Class Separation Delta

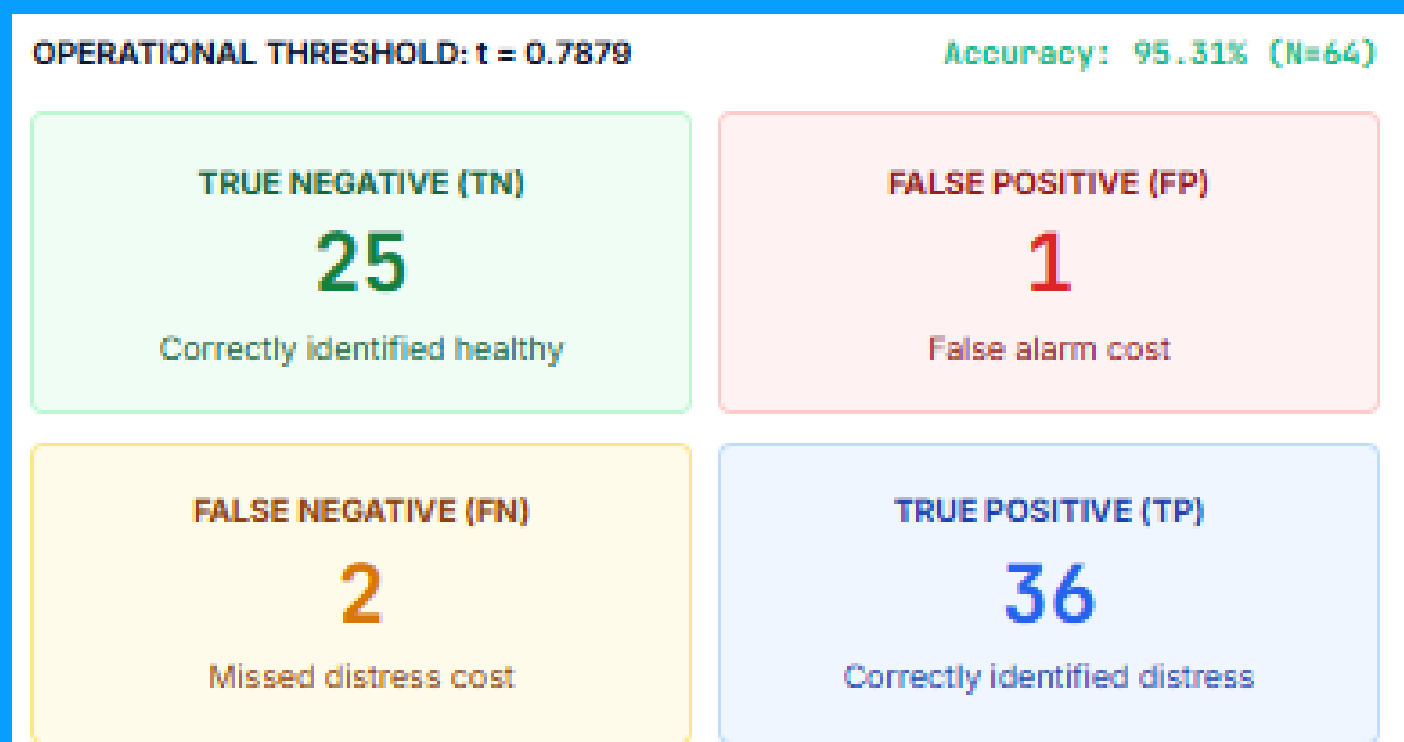
$\Delta = -55.7$ percentage points

FOUR FAMILY MODEL BENCHMARKING

Model Family	Test AUROC	Test AP	F1 Score	Cross-Company AP	Brier Score
👑 Random Forest (Champion)	0.9828	0.9908	0.9600 (@0.788)	0.9577	0.058
Logistic Regression	0.9828	0.9887	0.9143	0.9232	—
HistGradientBoosting	0.9767	0.9868	0.9067	—	—
MLP Classifier	0.9706	0.9819	—	0.7792 (-0.2623)	—

Selection Rationale: Random Forest selected as production champion with superior cross-company AP (0.9577), good calibration (Brier 0.058), and robust resistance to MLP cross-company collapse (-0.2623).

Confusion Matrix & Error Audit



CLASS PRECISION & RECALL

Healthy Class	P: 92.6% • R: 96.2%
Distress Class	P: 97.3% • R: 94.7%

INDIVIDUAL ERROR AUDIT

HD-2024Q2	FN • $p = 0.7835$ (marginal)
FIVE-2024Q3	FN • $p = 0.1745$ (structural)
DG-2025Q2	FP • $p = 0.7975$ (borderline)

Error Audit Finding: 2 of 3 test errors (HD-2024Q2 & DG-2025Q2) occurred within 0.01 of the decision boundary, confirming razor-sharp classification margin across the locked period.

COST & CALIBRATION OPTIMIZATION

PANEL A • ASYMMETRIC LOSS

BAYES DECISION CRITERION

C(FN) Miss Cost
5.0 x
Default penalty

C(FP) Alarm Cost
1.0 x
Audit cost

Bayes Cutoff p^*
0.1667
Theoretical

Bayes Risk Minimization: Theoretical threshold $p^* = C(\text{FP}) / [C(\text{FP}) + C(\text{FN})]$
 $= 1.0 / 6.0 = 0.1667$.

PANEL B • CALIBRATION ADJUSTMENT

OPERATIONAL THRESHOLDING

Tree ECE
 ≈ 0.109
Expected Cal Error

Theoretical Penalty
15 FP
False alarm surge

Operational Cutoff
0.7879
Pruned empirical

Operational Prudence: Empirical threshold $t = 0.7879$ trading off 2 FNs + 1 FP $\Rightarrow$ cost of 11
 $\Rightarrow$ not Bayes-optimal because RF's $\hat{p}$ is uncalibrated (ECE 0.109)

Theoretical Cutoff ($p^* = 0.1667$)

Operational Cutoff ($t = 0.7879$)

47-Feature Taxonomy & Ablation

BLOCK 1 • DOMINANT

ratios_latest

Current quarterly
solvency & liquidity

Δ AUROC -0.0172

BLOCK 2

ratios_yoy

Year-over-year
balance changes

≤ 0.003 drop

BLOCK 3

growth

Revenue & margin
expansion

≤ 0.003 drop

BLOCK 4

structure

Capital structure &
leverage

≤ 0.003 drop

BLOCK 5

stress

Interest coverage &
cash burn

≤ 0.003 drop

BLOCK 6

path

Rolling-window
trajectory

≤ 0.003 drop



KEY ABLATION INSIGHT

a decrease of **-0.0172 AUROC** (the largest among the 6 folds, but within the uncertainty of the 64 test samples)

PRACTICAL CONCLUSION

Latest-quarter solvency metrics carry virtually all actionable predictive information; historical multi-quarter trajectory features provide minimal marginal utility.

Entity Shortcut Leakage

TICKER MEMORIZATION

0.9858

Naive Ticker AUROC

DeLong test: $p = 0.7546$ vs Random Forest, proving headline AUROC can be matched via ticker memorization alone

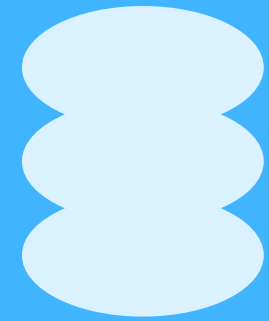
LEAVE-ONE-COMPANY-OUT

0.6280

Unseen Retailers AUROC

Severe degradation reveals that high in-sample test scores conceal an entity-identity shortcut rather than cross-firm mechanics.





The future of finance is intelligent, adaptive, and human-centered. AI is not replacing finance, it is redefining it.

**THANK YOU
FOR YOUR
ATTENTION**

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